

DAV Transport Deployment Guide

This guide describes how to:

1. Access your existing running dashboard from other devices (phones, tablets, laptops) on the same network.
2. Set up the system from scratch on a completely new server machine.
3. Configure the servers to run "forever" (24/7) and automatically launch whenever the computer reboots.

1. Accessing the Existing Dashboard on Other Devices

Since we have updated the frontend to resolve hostnames dynamically, you can access the dashboard on any device (phone, tablet, PC) connected to the same Wi-Fi or local network without setting up a new server.

Step 1: Find the Host PC's Local IP Address

1. On the computer running the server, click the Start menu, type Command Prompt (cmd), and open it.
2. Run the command:

```
cmd  
ipconfig
```

3. Locate the active network adapter (e.g., Wireless LAN adapter Wi-Fi or Ethernet adapter) and copy the IPv4 Address (typically looks like 192.168.1.XX or 10.0.0.XX).

Step 2: Open Inbound Ports in Firewall

To allow other devices to connect to your PC, make sure ports 3000 (frontend) and 5000 (backend API) are open:

1. Open the Start menu, type Windows Defender Firewall with Advanced Security, and open it.
2. Select Inbound Rules on the left, then click New Rule... on the right.
3. Select Port and click Next.
4. Choose TCP, specify Specific local ports as 3000, 5000, and click Next.
5. Select Allow the connection and click Next.
6. Keep all profiles checked (Domain, Private, Public) and click Next.
7. Name the rule (e.g., DAV Transport Ports) and click Finish.

Step 3: Access on Other Devices

Open the web browser on your phone, tablet, or another computer and go to:

`http://<YOUR_HOST_IP_ADDRESS>:3000`

(For example: `http://192.168.1.15:3000`)

2. Complete Guide to Setup the Server on a New System

To host this dashboard on a completely new machine, follow these steps to install the prerequisites and run the server.

Prerequisites to Install

Before downloading the code, install the following software on the new system:

1. Node.js (LTS Version): Download and install from nodejs.org. This installs both node and npm.
2. Git: Download and install from git-scm.com.
3. Python (version 3.10 or 3.11): Download and install from python.org. *Ensure you check the box to "Add Python to PATH" during installation.*
4. Tesseract OCR (Required for processing scanned image PDFs):
 - Windows: Download the installer from github.com/UB-Mannheim/tesseract/wiki. Install it and add the installation folder (usually C:\Program Files\Tesseract-OCR) to your system environment variables PATH.
 - Ubuntu/Linux: Run `sudo apt-get install -y tesseract-ocr`.

Step-by-Step Installation

Step 1: Clone the Repository

Open a terminal (or PowerShell on Windows) and run:

```
git clone https://github.com/dineshkumarjangid969-sketch/DAV-UPDATED.git
cd DAV-UPDATED
```

Step 2: Setup Environment Variables (`.env`)

1. Go into the backend directory:

```
bash
cd backend
```

2. Create a file named `.env` (you can copy `.env.example`).

3. Open `.env` in a text editor and fill in your details:

```
env
PORT=5000
NODE_ENV=production

# Email Scan Credentials (IMAP)
IMAP_USER=your_email@gmail.com
IMAP_PASSWORD=your_app_password
IMAP_HOST=imap.gmail.com
IMAP_PORT=993
IMAP_TLS=true

# Notification Sender (SMTP)
```

SMTP_HOST=smtp.gmail.com

SMTP_PORT=587

SMTP_USER=your_email@gmail.com

SMTP_PASS=your_app_password

Docling Service Configuration (if running AI document parsing)

DOCLING_URL=http://localhost:8000

> [!IMPORTANT]

> For Gmail, you cannot use your regular login password. You must go to your Google Account Settings -> Security, enable 2-Step Verification, and then create an App Password. Use that 16-character App Password as your IMAP_PASSWORD and SMTP_PASS.

Step 3: Install Node Packages & Build Frontend

1. Install Backend Dependencies:

bash

From the backend directory

npm install

2. Install & Build Frontend:

bash

cd ../frontend

npm install

npm run build

This creates a compiled production folder named build which contains the optimized React files.

3. Keep Server Active Forever (24/7 Autostart)

To ensure the backend and frontend servers remain running continuously and restart automatically if the computer crashes or reboots, we use PM2 (Process Manager 2) paired with a startup service hook.

Step 1: Install PM2 Globally

Open a Command Prompt or terminal with Administrator Privileges (Run as Administrator on Windows) and run:

npm install -g pm2

Step 2: Configure PM2 for Automatic Windows Startup

1. Install the Windows startup helper:

cmd

npm install -g pm2-windows-startup

2. Register PM2 as a background service:

```
cmd
pm2-startup install
```

(This ensures the PM2 daemon runs automatically when Windows boots up, before any user even logs in).

Step 3: Launch Dashboard Services using Ecosystem Configuration

We have already created a file named ecosystem.config.js in the root folder of the project. It launches both the backend and frontend services under PM2.

1. Navigate to the root directory of your cloned folder:

```
cmd
cd /path/to/DAV-TRANSPORT
```

2. Start the services:

```
cmd
pm2 start ecosystem.config.js
```

3. CRITICAL STEP: Save the running list of processes to PM2's database so it knows what to restore on system startup:

```
cmd
pm2 save
```

PM2 Cheat Sheet (Managing the Live Server)

- Check server statuses: pm2 status or pm2 list
- Check live terminal logs: pm2 logs (or pm2 logs dav-backend / pm2 logs dav-frontend)
- Restart the backend: pm2 restart dav-backend
- Restart the frontend: pm2 restart dav-frontend
- Stop the servers: pm2 stop all
- Update PM2 startup save point: Whenever you add new processes to PM2, run pm2 save to commit the configuration.